

Forensic Facial Comparison Report

AI-Assisted Similarity Analysis — Investigative Support Document

Report ID	03abec2e-2c31-4ae5-b680-6e5ff7c1c332
Case ID	DEMO-0001
Requesting Agency	Demo Agency
Created	2026-07-21T09:50:11.799205+00:00
Software Version	nextgen-forensics-scaffold-0.1.0

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1. Administrative Information

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Schema Version	1.0.0

2. Evidence Summary

Probe Image ID	probe.png
Probe Image Hash	b1efad8142d6bf44
Candidate Image ID	candidate.png
Candidate Image Hash	d88a05e0865d7201
Chain of Custody Notes	—

2.1 Purpose of Examination

Algorithmic comparison of a probe facial image against a candidate facial image to support investigative review. This document does not constitute a determination of identity.

3. Methodology

Processing followed a fixed pipeline: face detection, landmark localization, image quality assessment, anthropometric measurement, morphological feature comparison, embedding extraction (unmodified recognition model), and similarity scoring. Anthropometric measurement definitions follow the soft-biometric methodology of Tome et al. (2015), “Facial Soft Biometric Features for Forensic Face Recognition.” Landmarks were localized using the model specified in the appendix.

Clickable references:

- [Go to Landmarks Overlay \(Figure 3\)](#)
- [Go to Anthropometric Measurements Diagram \(Figure 4\)](#)
- [Go to Continuous Measurements Table \(Table 1\)](#)
- [Go to Morphological Variations Table \(Table 2\)](#)

4. Image Quality Assessment

Resolution	600x800
Blur Score (Laplacian variance)	102.8
Noise Estimate	0.28
Compression Artifact Score	0.074
Brightness	209.08
Contrast	14.96
Lighting Uniformity Variance	1.0
Pose Yaw	-2.5°
Pose Pitch	-90.0°
Pose Roll	-0.59°
Occlusion Score	0.091
Face Visibility	90.9%
Detection Confidence	0.909

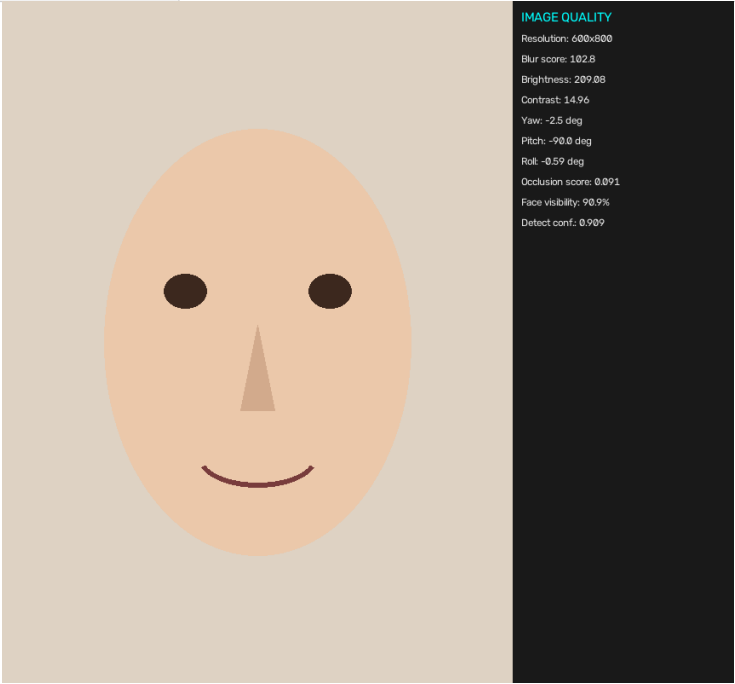


Figure 1. Quality visualization panel.

See Figure 8 for regional visual plots of the quality attributes.

5. Landmark Analysis

Landmark schema: 468 points. Model: mock-landmarker-68.

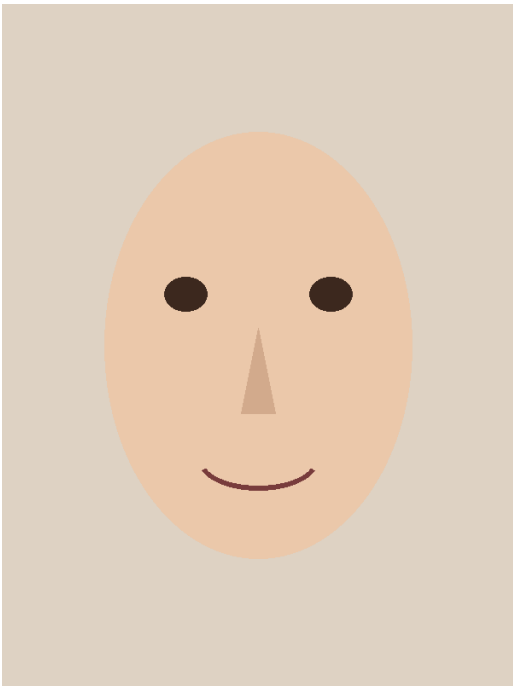


Figure 2. Original evidence image (probe).

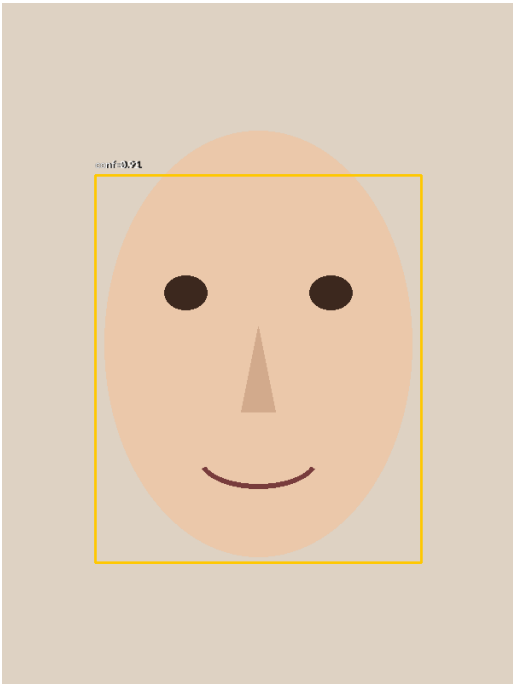


Figure 3. Detected bounding box.

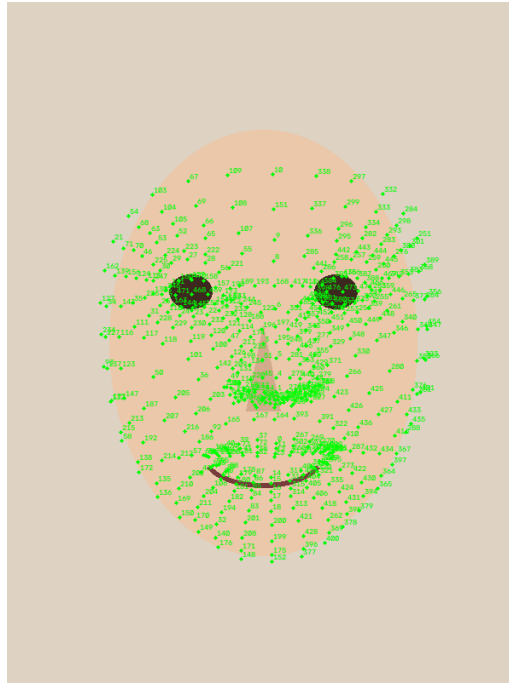


Figure 4. Landmark overlay, numbered.

6. Anthropometric Measurements

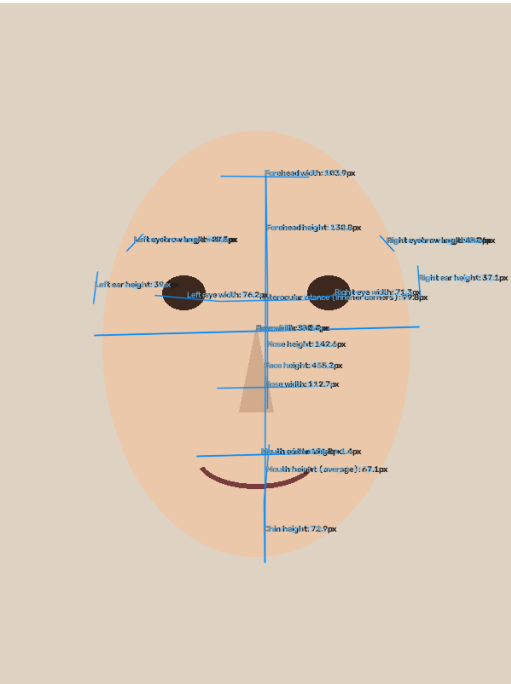


Figure 5. Anthropometric measurement overlay.

Feature	Region	Value	Unit	Landmarks
Face width	face	380.43	px	[234, 454]
Face height	face	455.16	px	[10, 152]
Interocular distance (inner corners)	eyes	99.84	px	[133, 362]
Right eye width	eyes	71.27	px	[362, 263]
Left eye width	eyes	76.21	px	[33, 133]
Right eyebrow length	eyebrows	24.55	px	[293, 300]
Left eyebrow length	eyebrows	27.62	px	[70, 63]
Right eyebrow angle	eyebrows	47.99	px	[293, 300]
Left eyebrow angle	eyebrows	-48.46	px	[70, 63]
Nose width	nose	112.73	px	[129, 358]
Nose height	nose	142.56	px	[168, 2]
Mouth width	mouth	151.75	px	[61, 291]
Mouth height (average)	mouth	67.08	px	[0, 17]
Mouth corner angle	mouth	-1.42	px	[61, 291]
Jaw width	jaw	380.43	px	[234, 454]
Chin height	chin	72.92	px	[152, 17]
Forehead height	forehead	130.82	px	[10, 168]
Forehead width	forehead	103.86	px	[109, 338]
Left ear height	ears	38.98	px	[127, 162]
Right ear height	ears	37.12	px	[356, 389]

Table 1. Continuous anthropometric measurements (probe image).

7. Morphological Comparison

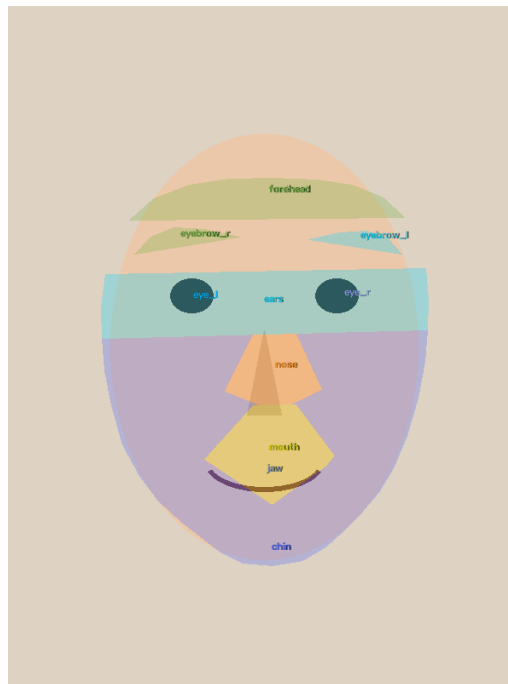


Figure 6. Morphological region segmentation.



Figure 7. Probe vs. candidate, matched landmark numbering.

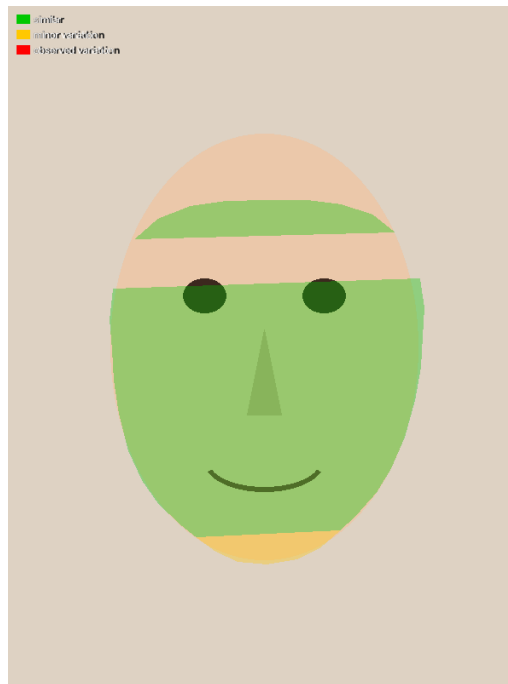


Figure 8. Difference visualization by region.

Region	Probe	Candidate	Comparison
face	wide	wide	similar
face	tall	tall	similar
eyes	average	large	minor_variation
eyes	large	large	similar
eyebrows	short	short	similar
eyebrows	short	short	similar
nose	average	large	minor_variation
nose	average	average	similar
mouth	average	average	similar
jaw	wide	wide	similar
chin	average	tall	minor_variation
forehead	average	average	similar
forehead	narrow	narrow	similar
ears	small	small	similar
ears	small	small	similar

Table 2. Morphological feature comparison, probe vs. candidate.

8. Similarity Analysis

Cosine similarity	-0.0207
Euclidean distance	1.4287
Angular distance (deg)	91.18
Threshold used	0.55
Decision category	below_threshold
Soft-feature Euclidean (supplementary)	3.944
Soft-feature Mahalanobis (supplementary)	3.255
Soft-feature Hamming (supplementary)	0.2

Plain-language notes: Cosine similarity measures how closely the two images' embedding directions align (1.0 = identical direction, 0 = unrelated). Euclidean and angular distance are alternative geometric measures of the same underlying comparison. The decision category reflects only whether the score crossed a configured threshold — it is not a determination of identity. The supplementary soft-feature distances are derived independently from this system's own anthropometric measurements and corroborate, but do not replace, the primary embedding-based score.

8.1 Observed Similarities

face, face, eyes, eyebrows, eyebrows, nose, mouth, jaw, forehead, forehead, ears, ears

8.2 Observed Differences

None at the configured category resolution.

9. Limitations

- No quality-based limitations were triggered by configured thresholds.
- No quality-based limitations were triggered by configured thresholds.

10. AI Findings Summary

The system computed a cosine similarity of -0.0207 against a configured threshold of 0.55, yielding a decision category of "below_threshold". This is a statistical similarity assessment, not a determination of identity.

11. Examiner Review

Examiner	_____
Signature	_____
Date	_____
Decision	☐ Confirmed ☐ Rejected ☐ Further review required
Comments	

12. Technical Appendix

Detection Model	mock-detector-v0
Recognition Model	mock-recognition-v0
Embedding Model	mock-recognition-v0
Similarity Algorithm	cosine_similarity
Landmark Model	mock-landmarker-68
Quality Model	opencv-laplacian-blur+geometric-pose-proxy
Hardware	n/a (scaffold demo)
Operating System	n/a
GPU	n/a
Software Version	nextgen-forensics-scaffold-0.1.0
Inference Time (ms)	367.64
Timestamp	2026-07-21T09:50:11.799205+00:00

13. Audit Log

Stage	Started	Completed	Status
detection_and_landmarks	2026-07-21T09:50:10.706758+00:00	2026-07-21T09:50:11.315708+00:00	ok
quality_assessment	2026-07-21T09:50:11.318695+00:00	2026-07-21T09:50:11.342510+00:00	ok
measurements	2026-07-21T09:50:11.342632+00:00	2026-07-21T09:50:11.361905+00:00	ok
embedding	2026-07-21T09:50:11.362094+00:00	2026-07-21T09:50:11.587954+00:00	ok
detection_and_landmarks	2026-07-21T09:50:11.588073+00:00	2026-07-21T09:50:11.605349+00:00	ok
quality_assessment	2026-07-21T09:50:11.607769+00:00	2026-07-21T09:50:11.633671+00:00	ok
measurements	2026-07-21T09:50:11.633800+00:00	2026-07-21T09:50:11.655245+00:00	ok
embedding	2026-07-21T09:50:11.655332+00:00	2026-07-21T09:50:11.797344+00:00	ok

14. Disclaimer

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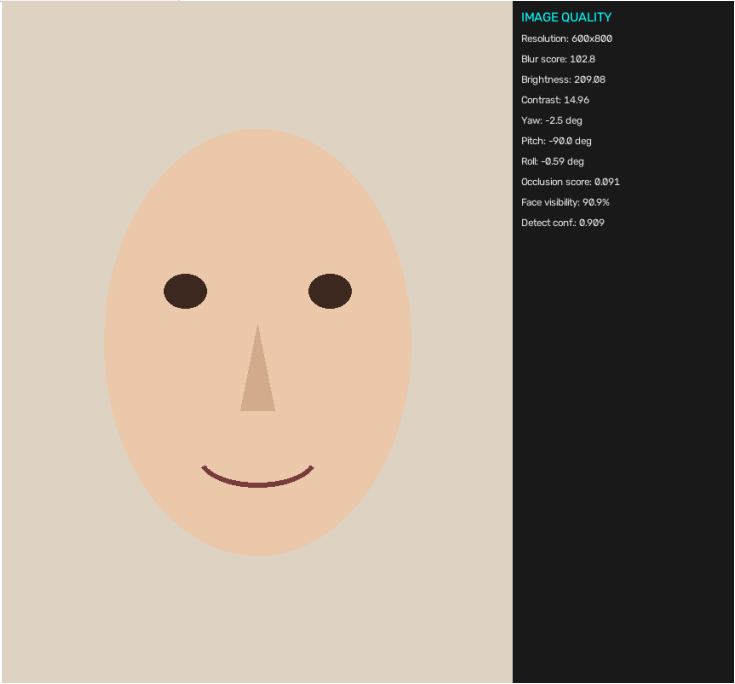


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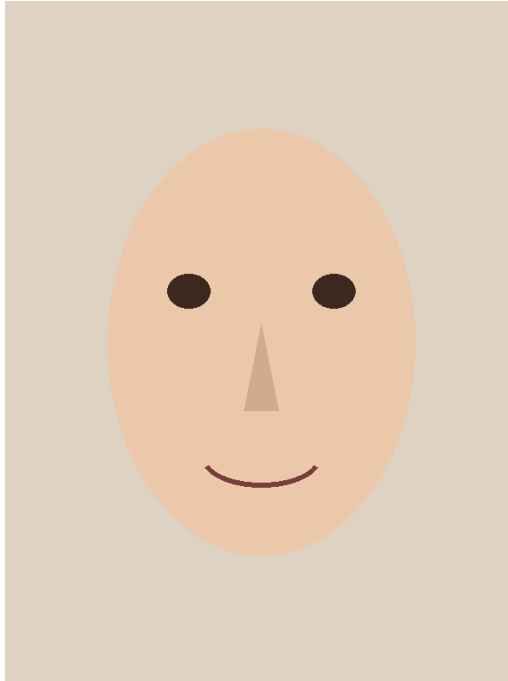


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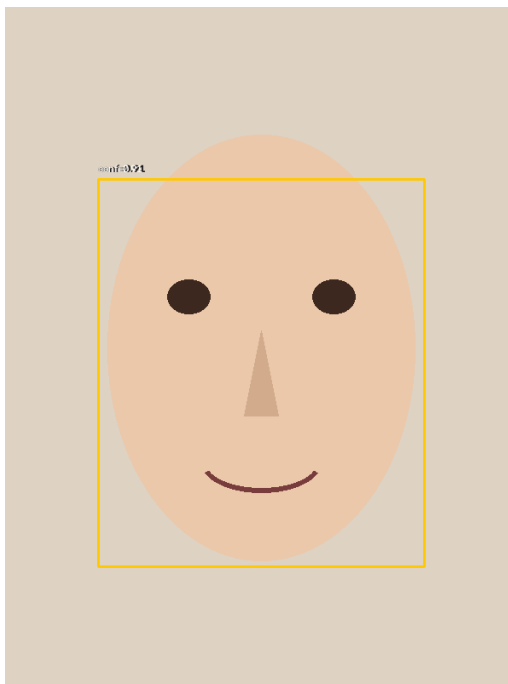


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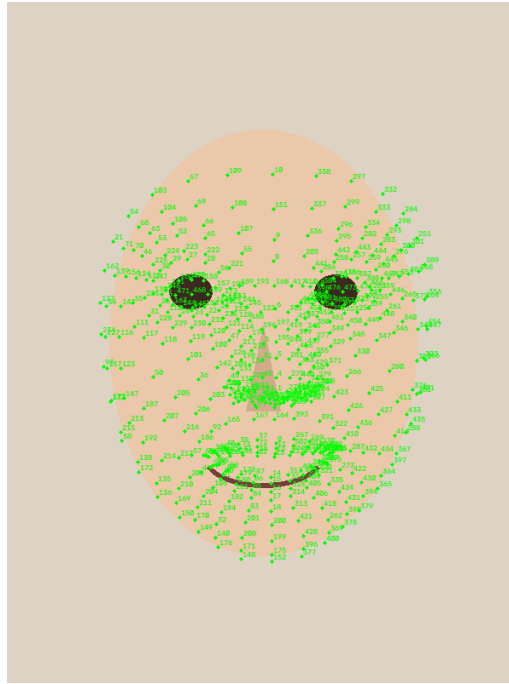


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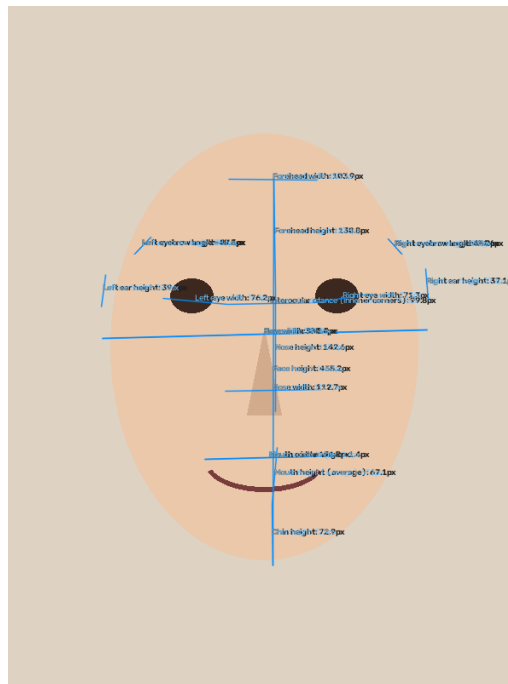


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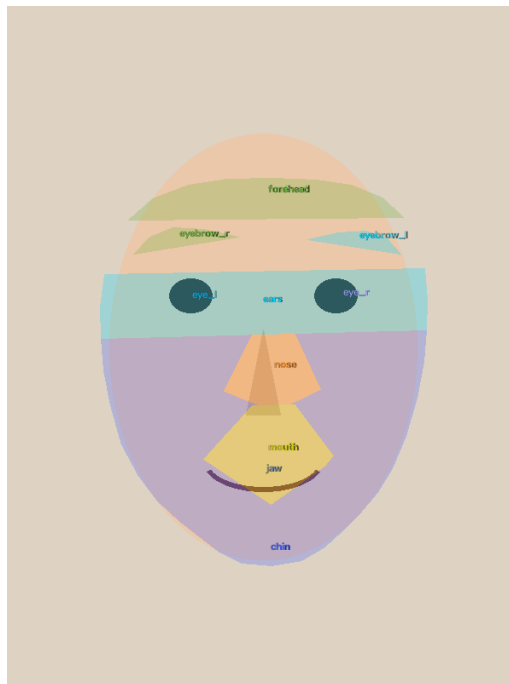


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embedding	2026-07-21T09:50:11.362094+00:00	2026-07-21T09:50:11.587954+00:00	ok
detection_and_landmarks	2026-07-21T09:50:11.588073+00:00	2026-07-21T09:50:11.605349+00:00	ok
quality_assessment	2026-07-21T09:50:11.607769+00:00	2026-07-21T09:50:11.633671+00:00	ok
measurements	2026-07-21T09:50:11.633800+00:00	2026-07-21T09:50:11.655245+00:00	ok
embedding	2026-07-21T09:50:11.655332+00:00	2026-07-21T09:50:11.797344+00:00	ok

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